

Wireless PA for Classrooms

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In large classrooms, many a times the teacher's voice is not audible to students in the back rows. So the teacher have to literally shout to be

heard by every student. Presented here is a circuit that can act as a wireless speech-aid for teachers so that their voice reaches every student even in a large classroom.

The circuit has two parts: wireless microphone and receiver amplifier. Wireless microphone is held by the teacher and whatever he speaks is transmitted over the FM band. Several receivers with speakers can be placed around the room to cover it completely.

Circuit and working

Fig. 1 shows the circuit of wireless microphone. The receiver amplifier circuit is shown in Fig. 2.

Wireless microphone section. The transmitter section of wireless PA for classrooms is built around two BC547 npn transistors (T1 and T2),

electret microphone (MIC) and a few other components. The audio signal is picked up by electret microphone MIC and amplified by transistor T1. The amplified output is fed to the base of transistor T2. Here transistor T2 modulates the L-C tank circuit comprising L1 and VC1 due to its varying junction capacitance corresponding to the input signal at the base. The tank circuit closely resembles a Colpitts oscillator. The 27pF capacitor connected with antenna isolates L-C circuit from the aerial capacitance. Inductor L1 is made by winding seven turns of 24 SWG copper wire around a 5mm diameter air core. For antenna, use any flexible wire.

Receiver amplifier section. The receiver section of wireless PA for classrooms is built around audio amplifier TDA2030 (IC1), FM receiver kit, loudspeaker (LS1) and a few discrete components. There are many FM receiver kits available with Sony ICs like CXA1019 and CAA1619. There are also some with Philips ICs. FM kits have the option to operate at either 6 V or 12 V. Select 12 V for this project and use a 12V, 1A adaptor for power supply. Select a kit that uses trimmers in place of a gang capacitor. Of course, even a ganged capacitor kit will do. But once the circuit is aligned with the FM transmitter/wireless microphone, it is difficult to fix the gang in a particular position.

Test Points

Test point	Details
Wireless microphone	
TP0	0V, GND
TP1	9V
TP2	Audio picked up by MIC
TP3	Amplified audio
Receiver amplifier	
TP0	0V, GND
TP1	12V
TP2	Audio output from FM kit
TP3	Amplified audio

PARTS LIST

Semiconductors:

- IC1 - TDA2030 audio amplifier
- T1, T2 - BC547 npn transistor
- D1, D2 - 1N4007 rectifier diode

Resistors (all 1/4-watt, 5% carbon, unless stated otherwise):

- R1 - 1-mega-ohm
- R2 - 10-kilo-ohm
- R3 - 47-kilo-ohm
- R4 - 470-ohm
- R5, R6, R8 - 100-kilo-ohm
- R7 - 4.7-kilo-ohm
- R9 - 150-kilo-ohm
- R10 - 1-ohm, 1W
- VR1 - 22-kilo-ohm preset
- POT1 - 22-kilo-ohm pot

Capacitors:

- C1, C6 - 22nF ceramic disk
- C2 - 100nF ceramic disk
- C3 - 1nF ceramic disk
- C4 - 4.7pF ceramic disk
- C5 - 27pF ceramic disk
- C7 - 1µF, 25V electrolytic
- C8, C9 - 22µF, 25V electrolytic
- C10, C13 - 0.1µF ceramic disk
- C11 - 2200µF, 25V electrolytic
- C12 - 100µF, 25V electrolytic
- VC1 - 47pF trimmer

Miscellaneous:

- MIC - Electret microphone
- CON1 - 2-pin connector
- CON2 - 3-pin connector
- BATT.1 - 9V battery
- L1 - 7 turns of 24 SWG wire over 5mm diameter air core
- ANT. - 18SWG wire antenna
- S1 - On/off switch
- LS1 - 4-ohm, 10W loudspeaker
- FM REC. KIT - FM receiver kit
- 12V, 1A power source

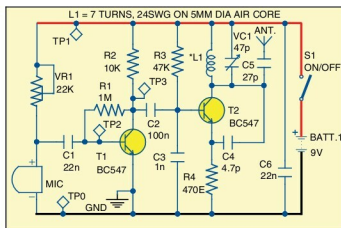


Fig. 1: Circuit of wireless microphone

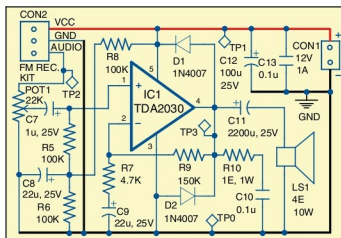


Fig. 2: Circuit of receiver amplifier section

